

GIOVANNI ALFADIANSYAH

Software Engineer | Full Stack Developer | Technical Consultant

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PROFESSIONAL SUMMARY

High-performing **Software Engineer & Technical Consultant** with 3+ years of professional experience in enterprise backend architecture, full-stack systems, and cybersecurity-compliant applications. Currently engineering robust, secure microservices for **Badan Siber dan Sandi Negara (BSSN)** utilizing **Java 21, Spring Boot, PostgreSQL, and SSO ConnectIDN (FAL2)**. Proven track record in modernizing legacy banking services to **Java & Spring Boot** for **Bank Mega (CCBM Division)**, architecting enterprise **Odoo ERP** systems for **Singapore Technologies (ST MES)** with **Python**, and developing full-stack portals using **C# (.NET), React.js, NestJS, and Docker**.

TECHNICAL COMPETENCIES

Programming Languages:	Java (21/17), Python, TypeScript, JavaScript (ES6+), C# (.NET), PHP, SQL, HTML5, CSS3
Backend Frameworks:	Spring Boot 3/4, Spring Security, NestJS, Node.js, ASP.NET Core, Laravel, Express.js, RESTful APIs
Security & Authentication:	SSO ConnectIDN (FAL2 / JWS / JWE), OAuth2/OIDC, TOTP MFA (RFC 6238), JWT (RTR), RFC 7807 Error Handling
Enterprise & ERP Systems:	Odoo ERP (Custom Modules, ORM, XML Views, Automated Server Actions), Manufacturing Execution Systems (MES)
Databases & Storage:	PostgreSQL, Redis, MySQL, Microsoft SQL Server, AWS S3, Flyway Migrations, Prisma, TypeORM
DevOps, Cloud & Testing:	Git, GitHub, Docker, Linux/VPS Administration, Testcontainers, JUnit 5, Postman, CI/CD, Agile/Scrum
Languages:	Indonesian (Native), English (Professional Working Proficiency)

PROFESSIONAL WORK EXPERIENCE

Badan Siber dan Sandi Negara (BSSN) Jakarta, Indonesia | Apr 2026 – Present
Software Engineer (Backend / Java)

- Cybersecurity-Grade LMS Backend:** Architect and develop high-security backend microservices for BSSN Learning Management System using **Java 21 and Spring Boot**, managing users, role hierarchies, dynamic course catalogs, and interactive evaluations.
- Advanced SSO & MFA Security:** Implemented enterprise authentication featuring **SSO ConnectIDN FAL2** (JWS signature verification & JWE payload decryption), **TOTP Multi-Factor Authentication (RFC 6238)** with emergency recovery codes, and Refresh Token Rotation (RTR).
- Database Architecture & Storage:** Engineered 24 JPA entities across 38 Flyway database migrations in **PostgreSQL**, integrated AWS S3 for multi-format learning media (PDF, MP4, Audio) with path-traversal sanitization.
- Automated Certification & Analytics:** Built dynamic HTML/CSS certificate generation with public QR Code verification, real-time learning progress heartbeat tracking, and an automated quiz evaluation engine supporting 5 question types.
- Testing & API Contracts:** Established comprehensive integration test coverage using **Testcontainers, JUnit 5, and Mockito**, standardizing OpenAPI/Swagger documentation and RFC 7807 ProblemDetail error handling.

PT. Wixander Beneficia Technology Jakarta, Indonesia | Aug 2025 – Mar 2026
Technical Business Analyst & Odoo ERP Consultant

- Enterprise ERP & MES Implementations:** Led full-lifecycle technical implementations of Odoo ERP solutions, notably for **Teck Leong Metals ERP** and **ST MES (Singapore Technologies - Manufacturing Execution System)**.
- Custom Backend Engineering:** Architected and developed custom Odoo Python modules, custom business logic, automated server actions, and XML view customizations across Sales, Purchase, Inventory, POS, Manufacturing (MRP), and Accounting.
- Shop-Floor Automation (Singapore Technologies):** Streamlined manufacturing operations for Singapore Technologies by integrating digital routing workflows, automated barcode scanning, and real-time inventory ledger tracking.
- Deployment & Version Control:** Supervised Git workflows, branch strategies, database migrations, and CI/CD deployments on Linux VPS staging and production environments.

PT Xsis Mitra Utama Jakarta, Indonesia | Oct 2023 – Apr 2025
Software Engineer (Backend & Full Stack)

- Client Placement: Bank Mega — Call Center Bank Mega (CCBM Division) [Contract: 3 + 12 Months]**
- Legacy Code Revamp & Migration (6 Months):** Spearheaded backend modernization of Call Center Bank Mega (CCBM) services, successfully migrating legacy **PHP** codebases to **Java & Spring Boot** within a 6-month delivery timeline.
- Production Maintenance & Operations (Post-Revamp):** Maintained, monitored, and enhanced CCBM production systems following the revamp, resolving critical incident tickets, optimizing database queries, and ensuring 99.9% uptime for call center agents.
- High-Performance RESTful APIs:** Designed and implemented scalable RESTful endpoints for real-time customer data retrieval and banking operations, achieving sub-second response times during peak call volumes.
- Internal Project: Pharmacy Management & E-Commerce System**
- Full-Stack C# (.NET) & React Development:** Co-engineered an internal pharmacy web platform using **C# (ASP.NET Core)** and **React.js** within a 4-member squad, automating prescription processing, catalog management, and stock reconciliation.
- Code Quality & Standards:** Enforced unit testing, conducted peer code reviews, and maintained comprehensive API contracts using Swagger and Postman.

Software Engineer Intern (Frontend / Full Stack)

- **Platform Rebranding & Modernization:** Spearheaded the complete frontend overhaul and rebranding using **React.js** and **Next.js**, boosting overall page load speed by 35% and significantly enhancing Core Web Vitals.
- **SEO & User Experience:** Implemented Server-Side Rendering (SSR) and dynamic metadata generation, improving search engine indexing and organic user acquisition.

PLABS.ID

West Java, Indonesia | Mar 2021 – Dec 2021

Software Engineer Intern (Full Stack)

- **E-Commerce Platform (Kapalasar):** Architected and delivered an end-to-end e-commerce platform using **Laravel (PHP)**, **Node.js**, and **React.js**, managing catalog management, cart logic, and transaction processing.
- **Database Schema & Query Optimization:** Designed normalized relational databases in MySQL, reducing query execution bottlenecks and improving system responsiveness.

KEY TECHNICAL PROJECTS

Online Tutoring & Student Management Platform (UP Learning Centre)

NestJS, TypeScript, PostgreSQL, Docker, Postman

Lead Backend Architect

- Engineered a production-ready backend API in **NestJS** with comprehensive JWT-based Role-Based Access Control (RBAC) supporting Admin, Teacher, and Student roles.
- Implemented complex scheduling algorithms to automate teacher allocation, recurring course bookings, and dynamic batch grouping.
- Designed normalized **PostgreSQL** relational schemas with indexing strategies, containerized microservices with **Docker**, and automated endpoint validation with Postman.

Centralized Faculty Monitoring & Analytics Portal — BEM KEMA FMIPA UNPAD

Laravel, PHP, MySQL, Bootstrap

Full Stack Developer

- Designed and deployed an integrated data monitoring dashboard for faculty leadership and student representatives, automating organizational performance reporting and reducing turnaround by over 50%.

EDUCATION & CERTIFICATIONS

Universitas Padjadjaran (UNPAD)

West Java, Indonesia | Aug 2017 – Aug 2023

Bachelor of Science in Computer Science (S1 Teknik Informatika / Ilmu Komputer)

- **Relevant Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Database Systems, Software Engineering, Computer Networks, Distributed Systems, Web Technologies.

Certifications & Intensive Bootcamps

Verified Credentials

- **Back End Development and APIs Certification** — freeCodeCamp (Node.js, Express, MongoDB, RESTful APIs)
- **Enterprise Software Engineering Bootcamp (.NET & Java)** — PT Xsis Mitra Utama (ASP.NET Core, C#, Java Spring Boot, Microservices)

LEADERSHIP & ORGANIZATIONAL EXPERIENCE

INSTAGRAM (Informatics Sports Arts & Games) — Project Officer

Jan 2019 – Apr 2019

- Led and orchestrated an executive committee of 30+ members to organize a faculty-wide multi-branch championship (6 sports, 4 arts competitions, 4 esports tournaments) with 500+ participants.

MIPA Awards — Vice Project Officer

Apr 2019 – Dec 2019

- Co-managed event budgeting, talent acquisition, cross-departmental operations, and sponsor relations for the annual FMIPA UNPAD awards gala.